Phase 3K | Air-brake integration

Pulsipher Farms / Moapa Valley Corn Maze

PF-TRAIN-S3K-R0 | 19 October 2026 UTC / 18 October in Nevada

This package develops the owner’s air-service and spring-parking direction into a functional trainwide architecture, documented component inquiries, calculated demand budgets and an instrumented verification plan. The drawings explain the connections and failure requirements. They are not a final plumbing diagram or fabrication release.

Recommendation	Reason and qualification
Automatic railway brake-pipe control	Develop a decreasing-pressure brake command with a local stored-air service response on every vehicle. A separate charging supply must not mask loss of the service command. Exact distributor, relay and brake-cylinder interface remain supplier decisions.
Separate spring-parking management	Retain spring-applied, air-released parking. Its sequencing, supervision and interaction with the service actuator require an integrated design; spring parking does not by itself establish an acceptable emergency stop.
Real semi-chamber candidates	MGM standard- and long-stroke combination chambers provide documented inquiry references. Match the exact force/stroke curves, mounting and brake mechanism before selecting a chamber or adapter.
One coordinated train	Include locomotive and all four cars. Verify empty and loaded braking, both sides of a separation, isolated-line failures, pressure loss, controls, heat and wheel adhesion.
First evidence step	Build and instrument a professionally designed brake bench, then validate every car and the full consist under an accepted test plan. No actual brake test or supplier application approval has occurred.

Preserved basis: one diesel locomotive, three regular coaches and one separate accessible coach; 36-inch gauge under evaluation; zero-percent longitudinal grade target. The proposed upper mass is 65,100 kg: 14,500 kg locomotive/hotel, three 13,200 kg regular coaches, and one 11,000 kg accessible coach. These are analytical cases, not rated gross masses.

01 | Control and connections

Revision PF-TRAIN-S3K-CONTROL-R0 • 19 October 2026 UTC / 18 October owner local date. **Preliminary functional requirements; not a pipework installation drawing, selected system, or operating procedure.** CR-G01 remains compressed-air service with spring-applied, air-released parking. All five vehicles require coordinated braking. The level-track target does not remove emergency, reserve, wheel-slide or parking requirements.

Recommended development direction

Develop **an automatic railway brake pipe (BP), a separate charging main (MR), and supervised parking/emergency permission**. A decreasing BP command makes each car's distributor use locally retained air to apply its service brakes. MR replenishes reserves independently. A local parking controller admits air to release the spring section only when the train is ready to move. The proposal keeps the owner's common combination-chamber preference open while using rail-appropriate train control. Matching a railway distributor to those chambers remains an engineering task.

The contrasting candidate is a positive-pressure semi-style service line, charging supply and independent emergency system. Its attraction is familiar components. Its key weakness is that a broken service line can remove the braking command while the intact supply keeps parking springs released. It therefore needs separately demonstrated fault detection and emergency actuation; repeating one trailer circuit four times is insufficient. Neither candidate has demonstrated lowest installed cost or accepted stopping performance.

Knorr-Bremse's current KE distributor/relay family offers graduated application and release; DB60 has direct release instead. Several KE variants exist, including passenger-coach applications and different backup versions [K-C01]. RDSO's actual LHB passenger-coach manual documents a separate charging pipe, decreasing-pressure BP, and KE-1/KR-1d equipment [K-C02]. Wabtec's SW4S-based FTSS3 is another documented inquiry family, although its cited application is freight [K-C03]. These establish real technology, not approvals for this train. Request a matched manufacturer system, including spring-brake integration, before selecting individual valves.

What connects to what

BP passes through the locomotive and four coaches and branches to each distributor and its reference-volume arrangement. MR branches through charging/check/protection functions into each local service reserve. The distributor commands a matched relay/load-control function; the reservoir supplies its energy; the relay supplies and exhausts the actuator's **service** side. A separate local parking/anti-compounding function supplies and exhausts the **spring-release** side. Both actuator sections ultimately load the same foundation brake.

A supervised energized-to-run safety function provides parking permission, emergency demand and traction inhibition. These are distinct functions, not one selected wire: normal pneumatic service or emergency can inhibit traction while spring-release permission remains valid. Loss of safety permission must cause a mechanically defaulting local BP emergency vent. It must not rely solely on a computer, communication network or remotely delivered command. The proposed parking default also applies springs, but its overlap with pneumatic braking requires engineered force arbitration. Shorted wires, bridged termination, welded contacts and spurious energization are explicit faults, not covered by merely choosing a normally energized circuit.

The topology JSON separates these functions; its arrows are not selected ports or instructions for assembling loose valves. Voltage, pressures, sizes, reservoir capacity, chamber count, response times and accepted failure tolerance remain open.

Required state and failure behavior

During normal service, springs remain air released while BP reduction meters pneumatic braking. During a BP-only rupture with MR intact, healthy local reserves must apply the service brakes. **The parking springs do not inherently apply in that event.** A separate parking command normally follows the stop. Small leakage that is masked by the charging feed must also be addressed; emergency venting must overcome or isolate charging.

MR rupture is different: BP may remain charged, so a check valve alone cannot command a stop. Supply/reserve supervision must initiate braking before the useful reserve is exhausted. Each reserve must resist back-drain through the broken main. Its usable energy must cover the accepted stop with measured leakage and application demand; stored air is finite.

Complete separation must stop **both portions**, at every coupling location. Both exposed BP ends must vent their connected vehicles. Self-sealing connectors or closed cocks can defeat that behavior. Bendix documents an SV-4 road-dolly function that intentionally avoids lead-unit emergency application after rear-trailer loss [K-C04]; copying that behavior would conflict with this project's requirement.

Power loss must remove traction permission and produce local pneumatic emergency action without software. Spring force may enter concurrently as its release air exhausts. Its timing, empty/loaded wheel adhesion and interaction with service pressure require proof; an arbitrary delay is not credited. Compressor loss initially leaves stored air available, but must cause a controlled stop before reserve readiness is lost.

A closed cock can leave a rear section falsely charged. A stuck relay can either prevent application or trap an applied brake. A broken linkage may defeat service and parking together even while pressure readings look normal. These conditions require actual actuator/brake-response verification, documented isolation status and a no-dispatch rule. Degraded passenger operation has not been authorized. Twenty explicit states are recorded in control_topology_R0.json.

Driver and connection requirements

Provide distinct train service, latched emergency and parking functions, with traction inhibition and deliberate reset. Indicate head/tail BP, charging pressure, each car's reserve readiness, service-delivery disagreement, parking state, isolation and manual-release status. A pressure lamp alone cannot prove brake torque or pad release. Loss or stale status must not silently become a ready indication.

For the future approved coupling procedure, the required sequence is: secure both portions independently; prevent movement; establish and inspect mechanical retention; connect correctly identified pneumatic and safety interfaces with people clear of pinch zones; establish the defined cock/termination configuration; charge while maintaining securement; then prove continuity, application, release and reserve readiness before removing restraint. For separation, first secure each resulting portion and establish parking; then isolate and depressurize the connectors by the accepted equipment-specific method before disconnection and mechanical uncoupling. **Valve order, pressure criteria and exact steps require the selected hardware manual and engineer-approved procedure.**

Reversal or towing must preserve every vehicle's brake control and both-halves separation response. Circuit continuity does not authorize pushing: coupler compression, visibility and movement rules remain separate reviews. Mechanical caging disables the affected spring function; it is a controlled rescue state requiring independent securement, tagging, restoration and retest, not a repair [K-C06].

Evidence needed to close this phase's decisions

Before hardware release, obtain exact OEM circuit/port drawings, force/stroke curves, relay and distributor response, allowable combined actuator force, reserve isolation behavior, load control and spring-transition logic. Bendix's multi-trailer guidance supports configuration-specific timing and end-of-line testing; its road regulations are not adopted here [K-C05].

Commissioning must record actual first/last-car pressure and mechanical responses during charge, graded service, emergency, release, parking and repeated applications. Inject BP-only rupture, MR-only rupture, every full-separation location, closed cock, wrong connection, power loss, failed relay, leakage and manual release. Define numerical acceptance limits before testing from the accepted stopping/adhesion analysis and OEM ratings. No test has yet been performed. Passing a schematic review does not establish a safe train.

Source URLs, retrieval dates, document locations and limits are in control_sources_R0.json; all safety-critical values remain unselected.

02 | Documented component candidates

PF-TRAIN-S3K-R0 | Research retrieved 19 October 2026 UTC / 18 October Nevada. Preliminary component comparison; no purchase or installation release.

Recommendation

Keep the requested compressed-air service brakes and spring-applied, air-released parking brakes. Develop two supply paths: a complete rail brake package from the bogie/brake supplier, and a properly matched mechanism using commercial-vehicle combination chambers. The latter retains the familiar semi-truck parts supply chain. A chamber alone does not specify the wheel brake, its heat capacity, or the train's stopping performance. No chamber, foundation brake, reservoir, hose diameter or operating pressure is selected by this register.

Verified combination-chamber candidates

Candidate	Manufacturer dimensions	Supply observation	Important distinction
MGM 3430051 / TR3030	2.50 in rated stroke; 8.00 in pushrod without yoke; 5/8 in rod; 3/8-18 NPTF ports; 5/8-11 studs, 1.540 in long	Current page advertises 2 days; minimum one. No stock confirmation or quote	Aftermarket S-cam drum actuator; rail adaptation unapproved
MGM 3230551 / TR3030LP3	3.00 in rated stroke; 12.00 in pushrod without yoke; same listed rod diameter, port and stud threads/length	Current page advertises 2 days; minimum one. No stock confirmation or quote	Extended-rod specialty variant; extra stroke is not a higher rail rating

Sources KC-01, KC-02.

MGM's 5019 brochure, PDF page 2, explicitly associates 3430051 with a 9.04 in casing length, 8.32 in service-case diameter, 8.21 in spring-case diameter, 4.75 in mounting-stud pitch, 5/8-18 UNF pushrod thread and 19.4 lb combination mass. These are nominal catalog dimensions, not a fabrication-tolerance drawing. The casing length excludes the projecting rod, studs, connections and maintenance tools. Sheet K02 uses a bounded envelope without inventing the internal casing geometry. KC-03

The currently linked service manual is EB 24-001 revision 02, November 2024. It identifies port 11 as service and port 12 as parking release, limits S-cam pushrod departure to ±3 degrees throughout travel, and requires a usable bottom drain orientation. The sealed spring section contains no user-serviceable parts. Maintenance access must accommodate inspection, hose removal and the manufacturer's release equipment; the crew must not open the sealed spring chamber. These constraints are incorporated without providing disassembly instructions. KC-04

Force and stroke evidence

Manufacturer literature contains a 100 psi standard/long-stroke comparison graph, demonstrating that output changes with stroke. It does not furnish an accepted current part-specific design curve or minimum spring force for our mechanism. An indexed MGM master-catalog link initially returned its index, then resolved to an invalid-file page on detailed retrieval; numerical force and air-volume tables could not be reverified. Values visible in third-party copies were deliberately excluded from the calculation basis. Service force, spring force, working pressure, release threshold, chamber air volume and environmental limits therefore remain null in the register. The model designation “30” is not treated as a constant 30-square-inch effective piston area. KC-05

Request the exact proposed assembly's pressure/force/stroke curves, minimum toleranced spring force, hysteresis, released air volume, temperature range and pressure limits. Repeat the match at new and worn friction material, hot and cold conditions, and maximum suspension/articulation travel. Rated stroke must not become the normal operating target. Road-service readjustment limits are maintenance context, not our rail acceptance limits.

Complete rail brake alternative

Knorr-Bremse's WheelAct Compact RZKT is a real pneumatic rail caliper family with optional spring parking. Its published family data include 50 kN total service-pad force, 45 kN parking-pad force identified with 4.8 bar, 305 mm width and 68 kg mass. TreadAct Compact instead combines a pneumatic cylinder, transmission and slack adjustment, with published family maxima of 60 kN service and 45 kN parking block force. These are references for supplier screening, not selected ratings for a 36-inch bogie; precise configurations, disc/wheel compatibility, release-pressure meaning and package drawings still require supplier confirmation. A complete rail package can eliminate much custom brake linkage, but size, price and delivery are unknown. KC-06

The same manufacturer page conflicts on lateral travel: its overview gives ±20 mm for all compact types, while detailed RZKT data gives ±25 mm. Both statements are retained in the source register. Accepted lateral movement remains unset pending a controlled exact-part drawing and supplier confirmation; neither value is used as an installation clearance.

Air supply and inter-car hardware

Candidate	Documented information	Evidence still required
Tramec Sloan TSJ95U tank	1,490 in³ / calculated 24.417 L; 9.5 in diameter × 27.5 in length; coated steel	Actual MAWP, certificate, mounting loads, fatigue, drain/relief arrangement and capacity selection
Phillips 12-006239 / 12-008239 gladhands	Blue service / red emergency; 1/2 in female pipe thread; keyed against cross-coupling	Flow, pressure limits, selected seals, locking, uncoupling and train emergency behavior
Phillips 11-8180-250 / 11-8190-250 hose	3/8 or 1/2 in ID; manufacturer lists 145 psi operating, 871 psi burst; −40 to 200°F	Approved hose-end combination, bend radius, fatigue, selected length and system pressure compatibility
Bendix compressor and air-treatment families	BA-921/BA-922 supply candidates; dryer removes contaminants before storage	Exact part, delivered flow versus speed/backpressure, duty, engine drive, lubrication/cooling, dryer capacity and purge budget

Sources KC-07, KC-08, KC-09, KC-10, KC-11.

The 145 psi hose limit is a concrete procurement constraint: burst pressure must never substitute for working pressure. A tank's dimensional listing similarly establishes neither its MAWP nor approval for the proposed system. Bendix brochure displacement CFM is not guaranteed delivered airflow. The compressor match needs the locomotive's idle speed and air demand while passengers board; low-speed charging and repeated service applications can govern.

Mechanical integration required before release

For a disc mechanism, establish torque from total pad normal force, accepted friction and effective disc radius; establish required actuator force through the actual linkage efficiency and motion ratio. For tread brakes, use the wheel supplier's approved block material and wheel thermal limits. Neither arrangement may inherit rubber-tire road adhesion. The bogie supplier must own torque reactions into its frame, brake attachment fatigue, axle/rotor connections, slack adjustment, retained pins and inspection access. Do not weld brackets to an accepted bogie without its reviewed attachment design.

Route flexible body-to-bogie hoses across the proven yaw/roll/jounce envelope. Inter-car hoses need supported loops, protected connectors and clearance from wheels, drawbars, steps and pinch points throughout curves and reversing. Function labels and physical keys are complementary. A shutoff/check valve can trap pressure and defeat intended pressure-loss detection; connector selection must follow the failure-state design. Verify both separated train portions, not only the last car. None of these catalog pages demonstrates assembled railway braking or jurisdictional compliance.

03 | Braking, air and thermal budgets

Preliminary calculations, October 18, 2026, America/Los_Angeles. No hardware selection or operating approval. The recovered Phase 3J design basis retains three regular coaches at 13,200 kg each, one accessible coach at 11,000 kg, and locomotive/hotel allowance 14,500 kg: **65,100 kg upper train study**. Nominal 54,755 kg and stress 59,840 kg remain separate budgets. These are not measured or rated masses. Track target stays **0% grade**; nonzero gradients below are off-design probes only.

Stop distance and response time

For initial speed v, net constant deceleration a, and an assumed interval t with no braking, **$s = v\,t + v^2/(2a)$**. The interval represents command-to-effective-braking delay; operator recognition/reaction is additional. No response time is claimed for the proposed pipe/valve system. Service buildup, emergency propagation, jerk and variable adhesion require a time-history model and tests.

Speed	Assumed a	0.5 s delay	1 s delay	2 s delay
5 mph	0.3 m/s²	9.444 m	10.562 m	12.797 m
5 mph	0.5 m/s²	6.114 m	7.231 m	9.467 m
5 mph	0.7 m/s²	4.686 m	5.804 m	8.039 m
7.5 mph	0.3 m/s²	20.412 m	22.088 m	25.441 m
7.5 mph	0.5 m/s²	12.918 m	14.594 m	17.947 m
7.5 mph	0.7 m/s²	9.706 m	11.382 m	14.735 m

None is an acceptance distance. At 7.5 mph, every extra second before braking adds **3.353 m** on level track. The engineer must set allowable response, stopping envelope, passenger deceleration/jerk and operating speed together.

Force, wheel torque and load path

At assumed 0.5 m/s², level-track translational contact demand is **$F = m\,a$**: train **32.550 kN**, each regular coach **6.600 kN**, accessible coach **5.500 kN**, locomotive/hotel **7.250 kN**. Coach plus locomotive demands sum exactly to the train demand.

For equal allocation among n braked axles, **$T_{axle} = F_{car}\,r_{wheel}/n$** before rotational inertia. With four braked axles and 0.30 m wheel radius, probes give **495 N·m per regular-coach axle** and **412.5 N·m per accessible-coach axle**. Equal left/right allocation halves these values per wheel. Four axles, wheel radii 0.25/0.30 m and rotor quantities are arithmetic inputs, not selected running gear. Two-axle cases are also tabulated. The 5% rotational allowance is a sensitivity, not measured inertia; actual unbraked-wheel inertia also changes contact-force distribution.

Foundation torque reacts through brake mount, bogie structure and the reviewed body/bogie connection. Longitudinal wheel/rail contact forces also create truck and body pitch reactions; a torque number alone does not size the bracket, axle, pin or weld.

Adhesion and empty/load balance

A simplified ceiling is **$a \le \mu\,g\,\beta$**, where β is the fraction of total normal load carried by braked contacts. At assumed μ = 0.05, braking all weight gives at most **0.490 m/s²**; braking half gives **0.245 m/s²**, before local unloading and dynamic effects. The 0.5 m/s² request already exceeds the first ceiling. The case array covers μ = 0.02/0.05/0.10/0.15 and β = 0.25/0.50/0.75/1. These are sensitivity values, not site adhesion measurements. RSSB documents that moisture, rust and grease can impair rail adhesion [K-N02].

An unchanged 6.6 kN regular-coach setting requires μ = **0.0510** at 13,200 kg but **0.0956** at the 7,040 kg tare allocation. A contact carrying 20% below its equal share raises the latter to **0.1195**. Thus fixed loaded-pressure settings can slide empty wheels. Load compensation, brake allocation and wheel-slide strategy need engineer/OEM review. Mixed-car cases calculate internal force imbalance but exclude coupler slack impacts; they are not coupler ratings.

Heat and parking

At 7.5 mph, E = ½mv² gives **365.903 kJ** for the upper train and **74.192 kJ** for one regular coach. Adding the separate 5% rotation probe gives **384.198 kJ** train energy. A 300-second repeated-stop interval produces only **1.220 kW** time-average train dissipation, while initial power during a 0.5 m/s² stop is **109.134 kW**. Average duty alone cannot establish rotor/drum capacity. Temperature, lining friction, heat partition, cooling, fade and emergency repetition remain unverified.

Parking demand before safety margin is **F_hold = mg sin(arctan(grade)) + F_external**. Level track removes gravity demand, not wind, residual traction, coupling disturbance or restraint requirements. Off-design grade and 1/5 kN external-force probes are not site wind loads or prescribed test forces. Required hold margin/duration, spring output at installed stroke, loss-of-air behavior and wheel adhesion remain held. Do not infer parking capacity from service-chamber size.

Chamber, disc and air calculations

Ideal service force **pA** is shown for assumed constant effective areas 16/20/24/30 in² and pressures 30/60/90 psi. For example 30 in² at 60 psi is **8.007 kN ideal**, not an MGM output rating. Diaphragm area changes with stroke; actual force, return spring, friction and installed stroke need OEM curves. MGM's long-stroke publication distinguishes stroke alternatives [K-N01]; it does not validate a constant-force design model or rail application.

For a two-face disc, **T = μ_lining (N_pad1 + N_pad2) r_effective**. With equal pads, each pad force N = L η F_chamber. Therefore T = 2 μ_lining L η F_chamber r_effective. If “clamp force” already means the sum of pad normal forces, do not multiply by two again. Lining μ, leverage, efficiency and radii in the inverse calculations remain unselected.

For initially atmospheric dead volume V_d and added swept volume V_s, isothermal air demand is **Δ(pV) = p_g V_d + (p_atm + p_g)V_s**. Reservoir pressure drop is Δ(pV)/V_reservoir. Using gauge pressure on swept volume alone misses air needed to occupy the growing chamber.

A purely illustrative 60 L reservoir, four actuators each with 0.25 L dead and 0.75 L swept volume, 2 L lines and 60 psig delivery consumes **27.497 L free air/application**, dropping **6.735 psi** each. Starting at 120 psig, five applications leave **86.326 psig**; six leave **79.591 psig**, below the arbitrary 80 psig study threshold. This is not a five-stop reserve approval. Actual volumes, pressure margins, leaks, spring releases, purge, thermal effects and failed-line isolation must replace the assumptions. Bendix confirms trailer valves perform distinct pressure-protection, anti-compounding and relay functions [K-N03]; reservoir volume alone does not prove emergency behavior.

Verification and next evidence

The reproducible Python script outputs **769 calculation records** with all inputs exposed. Fourteen checks cover mass/force equilibrium, analytical stopping benchmarks, numerical speed integration, disc force accounting and air conservation. These verify arithmetic only. Supplier force/stroke/volume curves, full pneumatic transient behavior, measured masses/axle loads, thermal duty and instrumented empty/loaded/failed-system tests are required before release. All twelve selection/approval fields remain blank.

04 | Installation interfaces and test release

The following records connect this brake phase to the frame, bogies, couplings and electrical system. A supplier part number alone does not close an interface. The responsible engineer must approve the configured system and evidence before construction or passenger operation.

Interface	Required evidence before release
Actuator to foundation brake	Actual service-force and spring-force envelopes versus stroke/pressure; lever or caliper ratio; motion and wear allowance; clevis/pin/rod geometry; allowable side load; environmental and rail-duty compatibility.
Brake to axle / bogie	Rotor/drum or tread geometry, friction data, thermal capacity, torque reaction, attachment fatigue, bearing heat and clearances throughout suspension travel and yaw. Final material, welds and holes are held.
Reservoir and compressor	Certified pressure vessel rating and volume, corrosion/drain access, protection and isolation, compressor free-air delivery at duty/temperature, dryer demand, recharge and leakage tests.
Inter-car services	Accepted coupler articulation and separation travel; hose bend radius and fitting pull loads; abrasion/pinch protection; distinctive positive identification; end-cock strategy and continuity test. Line count and electrical/pneumatic parking implementation remain functional study decisions.
Driver controls and electrical	Service command, emergency action and parking control; monitored permissives and diagnostics; loss-of-power behavior; propulsion inhibit; alarm interpretation; rescue/manual release procedure and restoration checks.
Frame and equipment zones	Exact mounting reactions, equipment mass/CG, service access and clearances. Revisit the Phase3J deeper crossmembers and unselected bogie interface before fixing brackets or line routing.

The controlled full-consist test matrix must include full separation at each of the four coupling locations, BP-only rupture, MR-only rupture, a closed or mispositioned cock, isolated positive-command-line failure in the road-style alternative, electrical loss and failed relay. Record both separated portions independently. K04 is a summary sheet; these cases remain separately required by the control-state matrix.

All line labels in the drawings describe functions. Actual valve port numbers apply only when quoted from the exact manufacturer document; a functional arrow is not an instruction to connect two catalog ports. Do not substitute self-sealing couplings that conceal a commanded brake-pipe vent without a designed and validated emergency strategy.

Verification stage	Record to obtain	Release owner
Component and bench	Exact part revisions; calibrated pressure, stroke and force/torque measurements; controlled charge, service, release, emergency and spring-parking transitions; leak and timing results.	Brake integrator / test engineer
Single vehicle	Empty/loaded load balance; all actuator stroke and wear margins; local response; brake drag and heat; parking hold; mechanical integrity and suspension/turning clearances.	Vehicle engineer / inspector
Full consist	Each car pressure and brake response; speed/deceleration/distance; repeated-stop temperatures; train compression/slack; both halves of separation and each isolated fault under an approved controlled test method.	Test director / operators
Operational release	Accepted stopping envelope and speed limits; startup/continuity checks; crew training; rescue/evacuation; maintenance limits and records; applicable authority and insurer requirements.	Responsible professionals / owner

05 | Deliverables, verification and next phase

The next supplier return should include the complete bogie/foundation-brake package and actuator match, followed by the pneumatic designer's valve/line schedule. Only then can mounting plates, hole patterns, welds, hose cut lengths and the final parts quantities be released. Catalog listing is the present availability evidence; no supplier has been contacted for this phase.

The analytical delay, deceleration, adhesion, chamber count, wheel radius, pressures and reservoir examples in this package are sensitivities. They are not acceptance limits. The test plan must define calibrated instruments, tolerances, safe test fixtures/containment, stop conditions and objective acceptance criteria before execution. No test is marked passed merely because a calculation or diagram exists.

Item	What it provides
Four engineering-development sheets	Functional trainline and failure connections; manufacturer-based actuator envelope; brake demand interfaces; instrumented test records. PDF/SVG/DXF, with units stated in the drawing README.
Interactive control explanation	Selectable operating and fault states, contrasting the recommended railway command with a road-style reference. This illustrates required behavior; it is not a pneumatic timing or safety simulation.
Editable calculations	Input and result JSON plus Python source for stopping, force, torque, adhesion, energy and air-use sensitivities. Assumptions and exclusions are preserved.
Component/source registers	Exact candidate identities, verified manufacturer dimensions, document dates and URLs; stock, ratings and application gaps remain explicit.
Release records	Failure matrix, supplier inputs and test requirements. Prior frame and connection packages remain unchanged.

Verification covers dimensional transcription, formula arithmetic and selected independent checks, source-to-drawing coordination, file integrity and visual inspection. It does not establish braking performance or professional approval. A supported browser interaction tool is unavailable here; JavaScript, data and static assets are checked, but live browser interaction is untested.

Phase 3L next develops running gear: bogies, wheel/rail geometry, axles, suspension and support footprints. It must incorporate this phase’s brake forces, actuator and foundation-brake packaging, heat, hose movement and reaction loads. Final control hardware and brake mechanical interfaces remain open until supported by actual configured supplier information and accepted analysis/testing.

06 | Linked control and calculation sources

Current primary-source links supporting Sections 01 and 03 are listed here. Component sources in Section 02 are linked inline. Full source records, document dates, locations and limitations are retained in the editable source archive. Retrieval was 19 October 2026 UTC / 18 October in Nevada unless a record states otherwise.

K-C01 | Knorr-Bremse Rail

Pneumatic Brake Control for Railway Applications

Portfolio evidence, not a dimensional/service manual or this-project application acceptance.

K-C02 | Indian Railways RDSO

Maintenance Manual of LHB Coaches, Chapter 3 Air Brake System

Real railway technical precedent. Its pressures, tanks, timing claims, components and dimensions are NOT adopted for this train or asserted applicable in Nevada.

K-C03 | Wabtec

FTSS3 pneumatic brake control

Alternative OEM inquiry family only; no documented compatibility with semi-truck chambers or small passenger cars.

K-C04 | Bendix

Trailer Valves

Road-trailer product descriptions. The last behavior is not the both-separated-portions emergency objective chosen here. Current service-sheet links returned no document.

K-C05 | Bendix

Selecting the Right Valve for Your Trailer Application

Road requirements discussed in the article are not an applicability decision for this attraction.

K-C06 | MGM Brakes

BrakeTECH Service Self-Study Training Program, Form 5043

Actuator concept and handling reference, not train torque/adhesion, stopping or rail approval.

K-N01 | MGM Brakes

Long Stroke Chambers for Drum Brakes, Form5031 revised7/2022

Manufacturer identifies 3-inch long-stroke S-cam families; no numerical curve values digitized or used. Not rail-application approval.

K-N02 | Rail Safety and Standards Board

Adhesion

Primary railway research summary on contamination/adhesion and mitigations. No project coefficient or code adoption inferred.

K-N03 | Bendix

Trailer Valves

Manufacturer descriptions of pressure protection, anti-compounding, service priority, relay and synchronization functions; no valve chosen or rail-compatible claim.